003-MOPR06 · DOCMERGE PRO · GOOGLE WORKSPACE MARKETPLACE

Architecture & Completion Status

What the product is, drawn as a picture; what is finished and proved; what is built but not yet proved; what is deliberately not built — and, for every one of those, the reason.

Field	Value
Document	claude_V001_MoPr06-docmerge_pro_gws_architecture_status_20260920_1815.pdf
Version	V001 — first issue
Date	20 September 2026, 18:15 BST
Author	Claude for Senthil / KP
Status	Released
Product version described	docmerge-pro-s1 v036 (code) · sprint tracker v036 · product workbook v047
Evidence basis	`npm run verify` from a wiped `node_modules` on 20 Sep 2026: 1,513 tests / 81 suites / 100% coverage, 25 contract tests, 4 boot tests, all gates green
Companion document	claude_V001_MoPr06-docmerge_pro_gws_manualtest_20260920_1815.xlsx — the manual test pack
Classification	Internal — contains no credentials and no customer data

1 · Document control

Version history, how this document is kept current, and who needs to see it.

1.1 Version history

Version	Date	Author	Change	Status
V001	2026-09-20 18:15	Claude for Senthil/KP	First issue. Created to answer a direct question: explain the product pictorially, and say what is complete, what is not, and why. Describes the state after external review PPLX V001 was incorporated and sprints S13–S19 were executed, including the adversarial verification pass that followed them.	Released
V002	—	—	PLANNED: re-issue after LC-1. The figures change shape at that point — the network boundary in Figure 1 gets crossed for the first time, and six amber boxes can become green or reveal defects. Do not re-issue before then; nothing else materially moves the picture.	Planned

1.2 How to keep this document honest

Every status in this pack is one of three things, and the distinction is the whole point of the document:

Status	What it means	What would be dishonest
VERIFIED	A command was run in a session and its output was observed. The command is named in `EXECUTION_EVIDENCE_S13_S19.md` in the code archive.	Calling something verified because its unit tests pass.
BUILT	The code exists and is unit-proved against recorded contracts — asserted URLs, headers and request bodies against a fake HTTP client. It has never called the live service.	Reporting this as done. Three documents did exactly that until a verification pass caught it (DM-F42).
NOT BUILT	No implementation exists, and a named decision or blocker says why.	Listing it as a gap without naming the decision that created it.

1.3 Distribution and approval

Role	Name	Purpose	Action
Owner / decision-maker	Senthil (KP)	Holds D-DM5 (pricing), D-DM7 (three ratifications), LC-1 and R-D22.	Read §4 and §5; act on Figure 3.
Reviewer (external)	Any cold reviewer	This pack is written to be read without prior context.	Challenge any VERIFIED badge by running the named command.
Engineering	Claude / future sessions	The single source for what remains.	Re-issue at V002 after LC-1, not before.

2 · Context — what this product is, and why this pack exists

2.1 The product in four sentences

DocMerge Pro generates documents from spreadsheet and form data inside Google Workspace. A customer builds a template, binds it to a Google Doc, and the product merges rows into copies of that Doc — substituting placeholders, inserting repeating table rows for things like invoice lines, swapping logo images, and exporting to PDF or DOCX before filing the result into a Drive folder.

Around that core loop sits the part that is actually unusual: a hash-chained tamper-evident audit ledger, enforced tenant isolation, a quota governor that refuses to promise throughput the platform forbids, and deletion with per-store row-count receipts. A mail-merge add-on has none of that.

2.2 Why this pack exists

On 16 September an independent external review executed the code rather than reading it. It found that the product was **domain-logic-complete, not code-complete** — roughly 11,500 lines of well-tested business logic in which every boundary to the outside world was an injected interface with no implementation behind it. `npm start` started nothing and exited zero. The migration script crashed on a dependency that was never declared. The supply-chain evidence covered 280 components, every one of them development-time.

All of that has been fixed and the fixes are proved by execution. This pack exists because "fixed" and "verified against Google" are different claims, and the difference is the single most important thing to understand about the product's current state.

2.3 How to read this pack

Section	Answers
Figure 1 (§3)	What are the parts, how does one request flow through them, and what has actually been proved at each layer?
Figure 2 (§4)	What changed between the two reviews? Read this if you want to know whether the work was a rewrite. It was not.
Figure 3 (§5)	What is blocked, on what, and who owns it? This is the action page.
§6	The completion table: every capability, its status, and the reason.
§7	What is deliberately not built, and the decision that says so.
§8	Open defects — the things known to be wrong right now.

> **The one sentence to take away.** The product now runs, serves, migrates and survives a restart — all proved by execution. Not one request has ever reached Google, Stripe or the Marketplace. Closing that last gap needs about an hour of the owner's time (LC-1), not another sprint of engineering.

2.4 Terms used

Term	Meaning
LC-1	Launch checklist item 1: create the GCP project and OAuth consent screen. The single external blocker.
D-DMn	An open decision owned by Senthil. D-DM5 is pricing; D-DM7 is three outstanding ratifications.
P-DM-nnn	A finding from external review PPLX V001 (16 Sep).
DM-Fnn	An internal defect. The canonical list is `DEFECT_REGISTER.md` in the code archive.
Adapter	The code that turns an internal call into a real HTTP request to Google, Stripe or Postgres.

3 · The product, and what has been proved at each layer

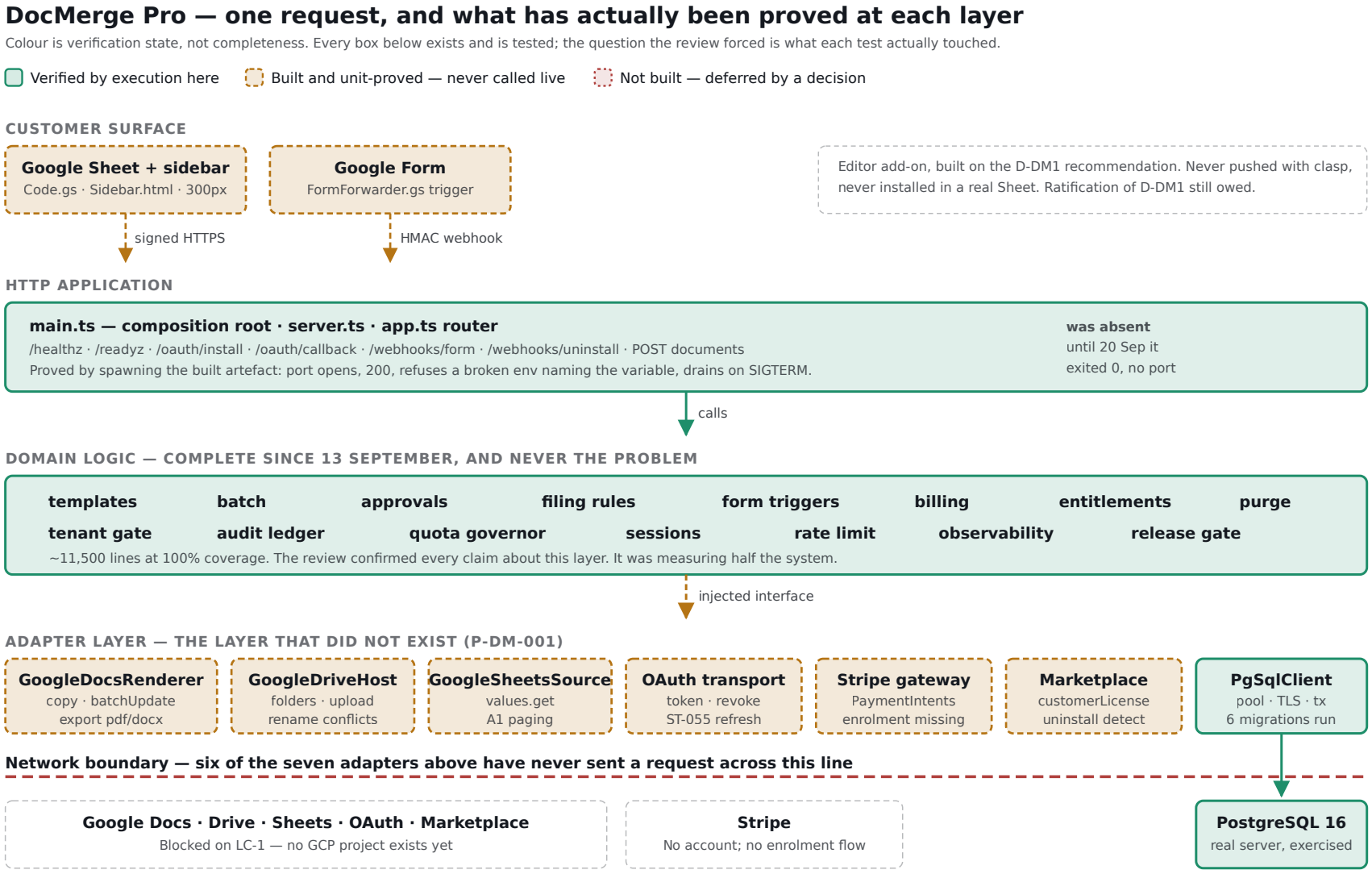


Figure 1 — one request, top to bottom. Green is proved by running a command and observing its output; amber exists and is unit-proved against recorded contracts but has never called the live service. The dashed red line is the network boundary — six of the seven adapters above it have never sent a request across it.

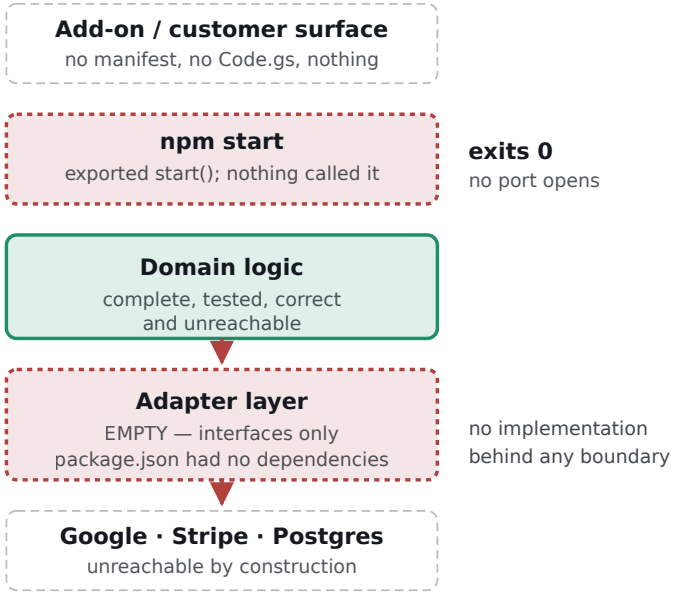
4 · What changed between the two external reviews

What changed between the two reviews — the same architecture, with the missing layer filled in

Both panels show an identical design. The difference is not a redesign; it is that four boxes which existed only as TypeScript interfaces now have code behind them.

16 September — as the external review found it

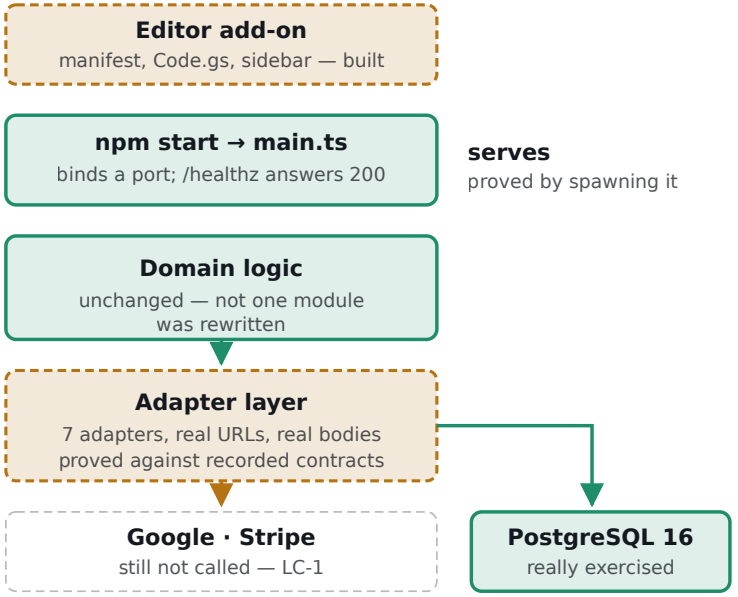
882 tests · 100% coverage · every gate green



Coverage could not see it: a file that was never written has no lines to leave uncovered — and the more boundaries you inject, the cleaner the number gets.

20 September — after S13–S19 and a verification pass

1,513 tests · 100% coverage · four new standing gates



The dependency injection that hid the gap is what made closing it additive rather than a rewrite: every interface an adapter needed was already written, already tested, and already correct.

Figure 2 — before and after. The two panels are the same architecture. Nothing was redesigned and not one domain module was rewritten; the work was additive, because every interface an adapter needed was already written, already tested and already correct. That is also why the gap was invisible: dependency injection is what made the code testable and what made the absence impossible to see, since a file that was never written has no lines to leave uncovered.

5 · What is blocked, on what, and who owns it

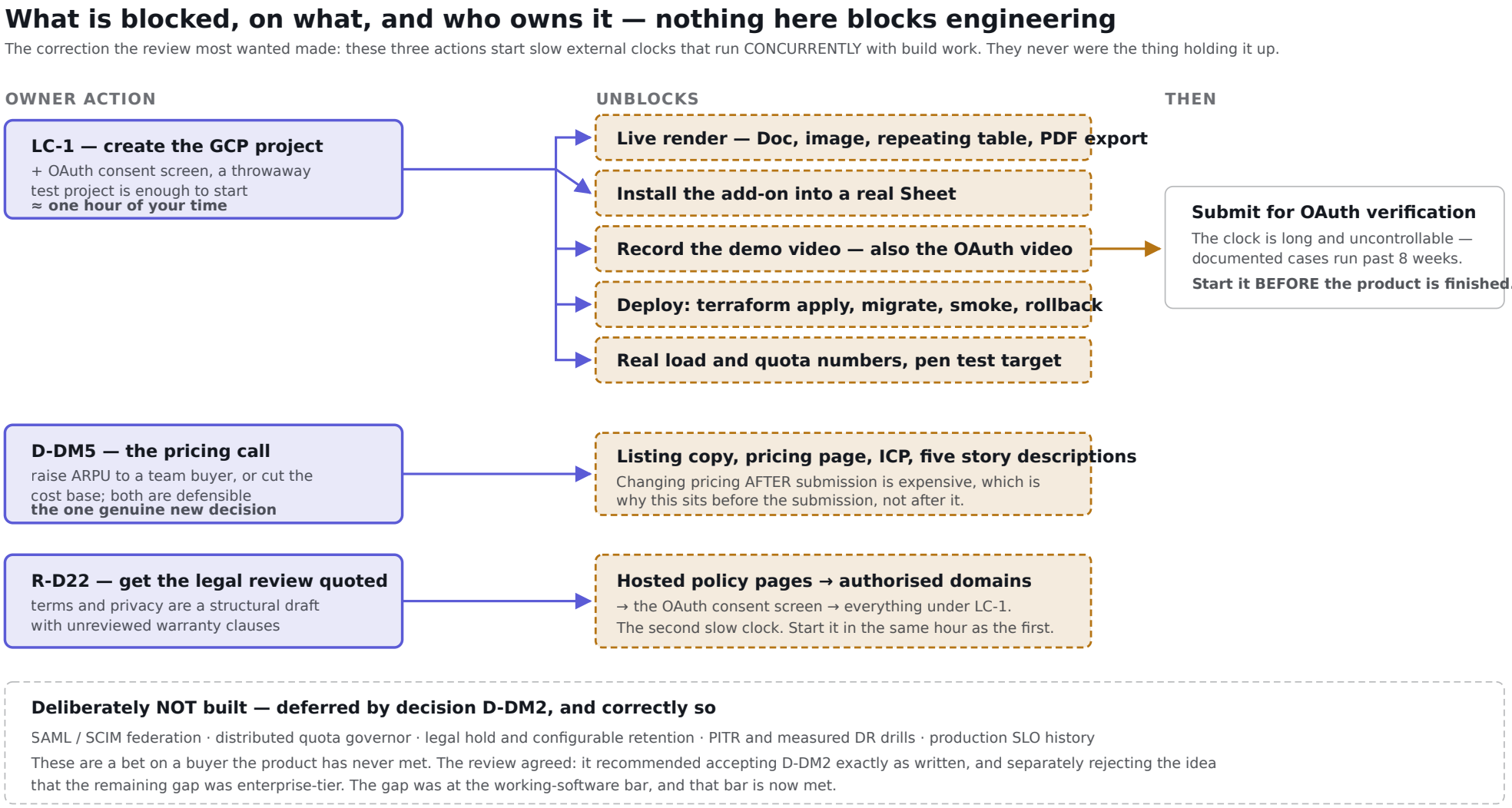


Figure 3 — the action page. Three owner actions, and what each one releases. The correction worth internalising: these do not unblock engineering. Seven sprints of engineering happened without them. They start slow external clocks — OAuth verification runs weeks to months — and those clocks can run at the same time as build work. The old reasoning had this backwards for three days.

6 · Completion table — what is done, and what it took to be able to say so

6.1 Verified by execution

Capability	Status	What exists	How it was proved / honest limit
Process & serving	VERIFIED	The built artefact binds a port, answers `/healthz` 200 and `/readyz` honestly, refuses a misconfigured environment naming the missing variable, and drains on SIGTERM.	`tests/boot.test.ts` spawns `node dist/main.js` — it does not import it. Re-run: `npm run test:boot`.
Database & migrations	VERIFIED	Six migrations apply against a real PostgreSQL 16 and a re-run is a clean no-op. Pooled driver with configurable size, statement timeout and TLS; transactions pinned to one connection.	Run against a live server in session, not an in-process engine. The earlier PGlite suite proved the SQL and bypassed the driver entirely.
Durability across restart	VERIFIED	Eight stores across seven new tables: webhook dedupe, the hash-chained audit ledger with its eviction anchor, approval requests, batch checkpoints, form triggers, rate-limit windows, sessions, deletion jobs.	Crash-restart invariant test per store plus PGlite contract tests. Honest limit: restart is simulated in-process, not `kill -9`.
Deletion actually deleting	VERIFIED	The published 30-day policy now has an executor that runs hourly, purges every registered store, and retains the job when a purge is only partial.	Found missing on 20 Sep by an adversarial pass (DM-F41) — the schedule was durable, the registry complete, and nothing called `due()`. Now contract-tested end to end.
Domain logic	VERIFIED	Templates, batch, approvals, filing rules, form triggers, billing, entitlements, tenant isolation, audit, quota governor, purge, release gate — ~11,500 lines.	Complete since 13 Sep and never the problem. The review confirmed every claim about this layer; not one module was rewritten during S13–S19.
Standing gates	VERIFIED	Eight gates run on every commit: tenant isolation, release integrity RG-01..06, runtime dependencies, boundary integration, published claims, contract suite, boot smoke, supply chain.	The integration gate fired 16 findings on its first run *after* the remediation, including two stubs written the same hour.
Supply chain	VERIFIED	13 real runtime components, `npm audit` clean at high severity, secret scan clean, SBOM asserted to be non-empty.	Before 20 Sep the SBOM's 280 components were entirely development-time, so 'audit clean' was a statement about Jest and TypeScript.
Backup & restore	VERIFIED	A drill executed against a real Postgres: 26 tables, 592 rows, restored and verified against the manifest. RTO 110ms.	Read with the caveat: 592 rows is a functional spread, not production volume, so 110ms is a floor. The first drill was re-run because it had covered a schema predating the durability work.

6 · Completion table (continued)

6.2 Built, unit-proved, and never called live — with the reason for each

Capability	Status	What exists	Why it is not verified
Google Docs rendering	BUILT	Copy the template Doc, one `documents.batchUpdate` carrying placeholder replacement, image swaps and repeating table rows, then export PDF or DOCX.	**LC-1.** No GCP project exists, so no request has been sent. The table row index arithmetic and section deletion across paragraph boundaries are the two things most likely to return a 400 on first contact.
Drive filing	BUILT	Folder creation, conflict handling including rename, multipart upload of exported bytes.	**LC-1.** Asserted against a fake HTTP client at the exact URL and multipart body. Never uploaded a byte.
Sheets reading	BUILT	Header-mapped row reads with A1 range paging, feeding the batch runner.	**LC-1.**
OAuth install & refresh	BUILT	Real POSTs to Google's token and revoke endpoints; refresh on the ST-055 path within 60 seconds of expiry, persisting a rotated refresh token.	**LC-1.** These URLs were string constants that no code had ever requested from S1 until S15.
Editor add-on	BUILT	`appsscript.json` generated from the audited scope set with a drift test, `Code.gs`, a 300px `Sidebar.html`, and the sidebar host shim the code assumed since S3.	**LC-1 + D-DM1.** Never pushed with `clasp`, never installed in a real Sheet. Built on the review's recommendation; the decision itself is still owed.
Uninstall detection	BUILT	Polls the Marketplace `customerLicense` endpoint, because the Marketplace pushes no uninstall webhook (VER-DM1).	**LC-1.** Needs the published listing's application id. Found missing by the integration gate — it had been an interface with nothing behind it.
Billing / Stripe	BUILT	Real PaymentIntents with an idempotency key derived from account and description; a charge routed to the wrong provider is a configuration error, never a silent decline.	**No Stripe account, and the enrolment flow is not built.** Nothing writes the customer id yet, so an unenrolled account is refused and logged rather than charged.
Deployment	BUILT	Dockerfile, Terraform for Cloud Run and Cloud SQL, a deploy workflow that fails closed naming LC-1 rather than reporting a false green, and a synthetic monitor that has been run for real locally.	**LC-1 + no Docker daemon here.** Terraform has never been `init`-ed; expect the first `validate` to surface a few provider argument names.

> **Why this whole block shares one reason.** Six of these eight wait on the same hour of work: creating a GCP project and consent screen. A throwaway test project with a single test user is enough to start — the verified production project is a later step. That is why Figure 3 puts LC-1 at the top and why the review called the old sequencing backwards.

7 · Deliberately not built — and the decision that says so

A gap with no named decision behind it is a defect. Each of these has one.

Capability	Status	Decision	The reasoning, kept so it can be rejected on its merits
SAML / SCIM federation, domain policy, break-glass	NOT BUILT	D-DM2 — launch first	A reviewer scoring against an enterprise bar always returns 'not enterprise grade' — that is a verdict on a bar nobody chose. The external review examined this and agreed, recommending D-DM2 be accepted exactly as written, while separately rejecting the idea that the remaining gap was enterprise-tier. It was at the working-software bar, and that bar is now met. Trigger: the first written SSO request from a real buyer.
Distributed quota governor	NOT BUILT	D-DM2 / RR-06	Today's governor paces a single process, which is honest for a single process. Distributing it is work for a scale the product has never seen.
Legal hold, configurable retention, export evidence	NOT BUILT	D-DM2 / RR-08	The purge foundation exists and has row-count receipts. Legal hold is a feature for a buyer the product has not met.
PITR and measured DR drills	NOT BUILT	D-DM2 / RR-09 — and partly LC-1	A drill harness exists and one local drill has run. PITR needs real managed infrastructure.
Production SLO history	NOT BUILT	D-DM2 / RR-10 + LC-1	Error budgets and a synthetic monitor exist. History cannot be manufactured; it needs 7–30 days of a running staging environment.
Real e-signature	NOT BUILT	D-DM6 — open	The feature was renamed to Approval requests because it produced no signature artefact, no binding of signer intent to a document hash, no signer authentication and no consent capture. The legal exposure is closed by the rename and a gate that fails the build on any signature claim. Integrating a provider is now optional and cheap to defer until a customer asks.
Discovery interviews	NOT BUILT	D13 — locked, portfolio-wide	For a self-serve listing the first fifty installs are cheaper and more honest evidence than twenty interviews. The telemetry to capture that is built.
CASA Tier 2 assessment	NOT BUILT	VER-DM5 — likely not required	CASA is triggered by *restricted* scopes and the product requests none. Worth noting this also deleted one of the three original differentiators: the listing can no longer claim a CASA badge, because there is none to earn.
Third-party penetration test	NOT BUILT	ST-068 — needs budget and a target	The scoping brief and vendor outreach are written and ready to send. It needs a deployed environment to test, which is LC-1, and a budget line.

8 · Open defects — what is known to be wrong right now

Seven open, none of them blocking. The canonical list is `DEFECT_REGISTER.md` in the code archive; these are the ones a reader of this pack should know about.

Id	Defect	Why it is not fixed yet	Impact if left
DM-F27	`AdminAuditLog.exportCsv` omits the previous digest, so the audit CSV a customer receives cannot be verified by them — the chain links are missing from the artefact whose whole value is the links.	A new verifiable exporter supersedes it. Changing the original alters an existing export's output, which is the owner's call.	A customer who asks to verify their own audit trail cannot, using the original export.
DM-F29	An expiry observed in the approval sidebar is never written back to the stored request; only the next action flips it.	Needs a service method that expires the stored request rather than a copy.	A request displays as expired and is still actionable until someone touches it.
DM-F30	Two gallery rules for the payment-reminder template are not mutually exclusive, so past 30 days both the gentle and the firm paragraph render.	The rule engine has no `less_than` or negation operator; a proper fix needs a new operator rather than a patched fixture.	One of twenty launch templates produces contradictory copy at a specific data boundary.
DM-F31	No install handoff from the OAuth callback to the sidebar — the callback returns a session token as JSON and the user must paste it.	Needs a redirect landing page or a short-lived pairing code.	Blocks the demo recording, which is also the OAuth verification video. Fix before recording.
DM-F37	`WEBHOOK_SECRET` is one global secret, handed to customers on the standalone form-forwarder path.	Per-account webhook secrets are owed before that path is offered publicly. The add-on path avoids it.	Do not publish the standalone forwarder path until this is done.
DM-F38	The application and the migration job share one Postgres role.	Owed with the first real deploy; the GCP identities are already split.	Least-privilege gap at deploy time.
DM-F44	No composed path drives dunning — the notifier exists and nothing calls it.	Same shape as DM-F41: it needs a route or a worker, not a boundary.	Failed-payment emails cannot be sent in production. Matters only once there are payments.

> **Why DM-F41 and DM-F44 are the interesting pair.** Both are the same class: a correct mechanism that nothing invokes. The integration gate built in S19 asks whether a boundary *has* an implementation; it cannot ask whether anything *calls* it. That gap is now the most valuable open question for the rest of the portfolio, and the question that finds it is simply: *what runs the loop?*